

“Exploring NFL Statistics”

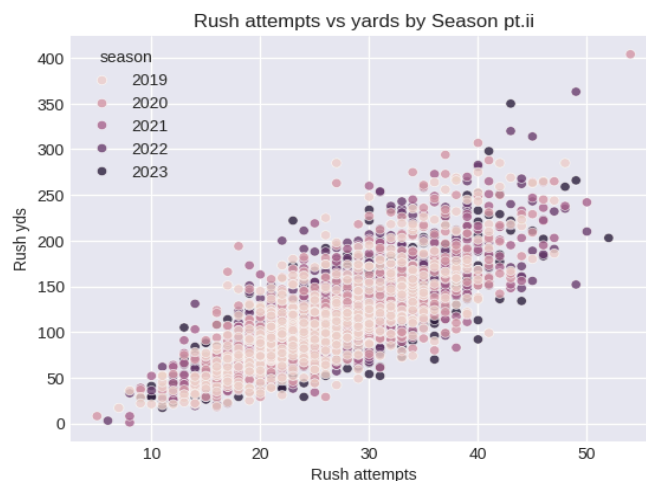
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This project will be diving into basic NFL statistics for analysis. For this Project, I will be working with datasets that contain statistics from previous NFL seasons, specifically including datasets that contain information on NFL drives, player and team stats, scoring data, and seasonal data.

My primary focus for this project is to explore NFL data, focused on passing and rushing statistics from the last 5 seasons. The datasets I will be using will be the last 5 years of data from each of the datasets I mentioned above. (Each year is its own dataset, so I could be working with up to 20 total individual datasets.) In my analysis, I will aim to answer some questions, such as: How has the league changed in 5 years? How have the team stats evolved? How have individual team stats evolved? Is there any correlation between the passing game and the rushing game? Is there any significant difference in the data from each game across the last 5 seasons? Which team or teams have changed the most in the last 5 seasons? What does the scoring data have to say about the overall offense and defense of the teams in the league? Who are the best passing and rushing teams in the past 5 seasons?

In order to find answers to my questions, I first ingested all the code I would need into a Jupyter notebook. From the ‘stats’ files, I concatenated each file and created one giant dataframe to include all the stats from 2019 to 2023 to act as my main dataframe. From that, I separated the dataframe into ‘Rushing’ and ‘Passing’ dataframes, with each one containing its respective stats. I created efficiency variables such as the rush average and pass average variables to their respective dataframes, and also a rush success variable to the rushing dataframes as a counterpart to the passer rating variable. I also created variables that held the totals and means of each column in each of the dataframes and used those variables to create several tables.

For my analysis, I started with the league-wide rushing data before looking at the team rushing data. In my analysis, I looked at scatterplots showing the relationship between the number of rushing attempts vs. number of yards gained by year and the total number of touchdowns scored by season. From the visualizations, you can see that there is not a lot of variation from season to season. The distributions are very similar, and there is a lot of overlapping data. From this, we can answer one of the questions I had and conclude that the rushing game did not change very much, at least in the span of these 5 seasons.

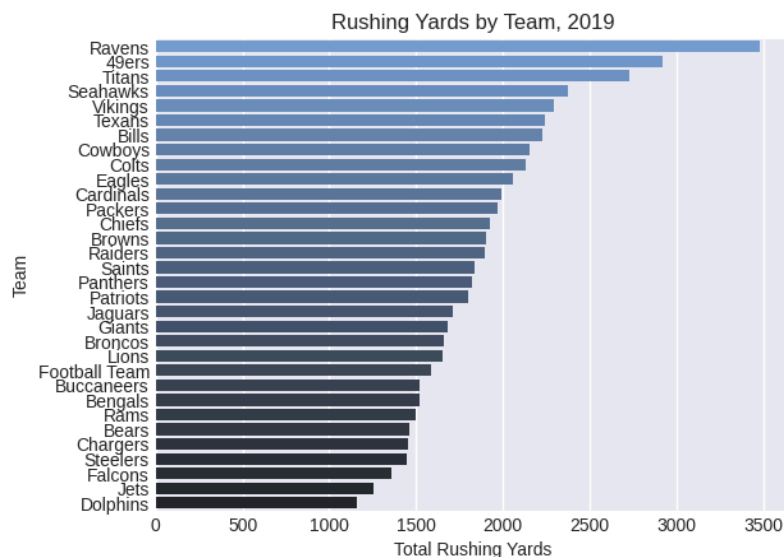


This conclusion is supported by the slope and R^2 values. Each season has an R^2 of around .60, ranging from .57 to .62. This means that about %60 of the rushing yards is explained by the number of rushing attempts. The slopes were also similar, showing that each season had an average of about 5.3 -5.4 yards per attempt.

While it appears that there wasn't a great deal of variation in the league data, the same cannot be said for the team data. In my analysis, I looked at box plots that displayed the distribution of rushing

attempts per team and barplots that displayed total rushing yards per team, and came to several conclusions: The Ravens are always in the top five and are the #1 rushing team 3 out of the five years. Based on the data, we can see that the Ravens are undoubtedly the best rushing team of this era. Their record-breaking year had them almost reach 3500 rushing yards, and they did it again the following season as well. (Also important to mention that Lamar Jackson broke the single-season rushing record for a QB in that 2019 season as well.)

Another team with rushing success is the Titans, who were top 3 in total rushing yards from 2019 to 2021, before falling to mediocrity the next two seasons. Their success was largely due to Derrick Henry, who led the league in rushing in 2019 and 2020, before getting injured halfway through the 2021 season. But what's crazy is, Henry had nearly eclipsed 1000 yards (the benchmark for rushing yards) in only 8 games (!!), leading the league in yds/game for a 3rd straight year and leading the Titans to 2nd in the league for total team rushing yards despite playing only half the season. It is interesting to note, though, that despite having a fully healthy Derrick Henry for the next two seasons, and him having the second most yards in both seasons, the Titans only ranked league average for total team rushing yards, which goes to show that one superstar running back is not all you need for team success.



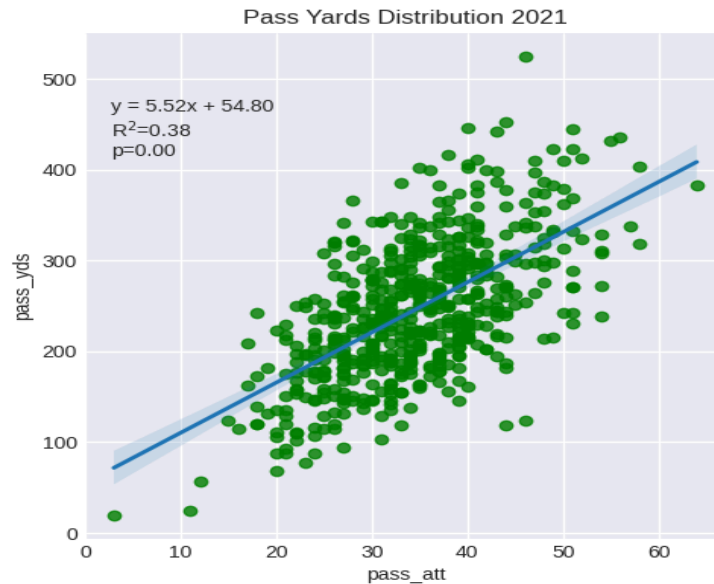
In addition to rushing yards, I also looked at rushing success using the variable I mention before. The `rush_success` metric is a better metric to determine who was actually the best and worst rushing teams, per average, in each season and by looking at both the sets of graphs showing rush success and the graphs displaying the rushing totals, we can come to some interesting conclusions:

- The Ravens, who I said before was undoubtedly the best rushing team in this time period, had by far the best rushing success in 2019 and 2020 as expected, but they were outside of the top three in 2023, despite leading the league in total rushing yards. In 2021, they were 5th in total rushing and outside of the top 5 in rushing success, likely due to Lamar Jackson being injured a handful of games that season.

- The Buccaneers, who I mentioned were bottom 10 in total rushing yards in each of the 5 seasons, actually were league average in rushing success in 2021, before plummeting to the worst rushing team in the league the following season. It is also interesting to note that this team won 11 games and won the superbowl in 2020, with a bottom 10 rushing team, and won 13 games in 2021 with a league-average rushing team. In 2022, however, they only won 8 games (out of 17) and had the worst team rushing success this season. Tom Brady was the QB for all three of these teams and put up pretty insane numbers for a 40+ year old, something we will see when we look at the passing stats.

To summarize this first part, I found little variation in the league-wide rushing data, supported by the slope and R^2 values, before looking at the team data and seeing far more inconsistencies, with teams like the Ravens standing out. Now we will do the same analysis with the passing game.

The group of scatterplots showing passing attempts v.s yards shows a similar idea to the rushing graphs in that there isn't too much difference in variation from year to year. Each plot has a positive trend and shows relatively similar R^2 values and slopes. The difference is that there is more variation in the passing data than the rushing data, as the R^2 values range from 0.32 to 0.47, a spread that is larger than the spread of the rushing data by 0.1. The average R^2 value for the passing data is around 0.4, and the average for the rushing is 0.59, which confirms that, on average, the variation in the passing is significantly greater than the variation in the rushing.

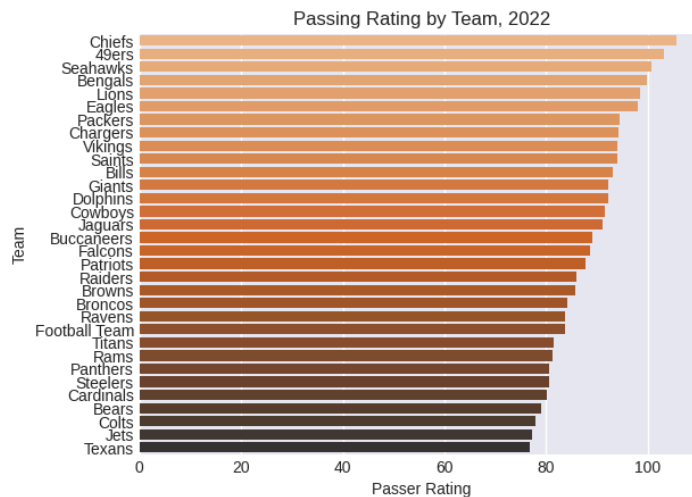


Just like in the rushing data, the slope is also similar across seasons, and just like in the rushing data, the range of the slope is similar, with both the slopes for the rushing and passing data having a range of about 0.5. The difference, however, is that the passing data has a slightly higher average slope at 5.67 compared to the rushing, with an average of 5.21. In context, a higher slope means, on average, every single increase in pass attempts is going to yield a higher increase in yards compared to a single increase in rushing attempts.

When I looked at the bar charts displaying the passing yards by team, the first thing that jumped out at me was how dominant the Chiefs were in their 5-year span, leading the league in total passing 3 out of the 5 years and not ranking lower than 2nd, which is remarkable. We talked about how dominant the Ravens were at this time in the running game, but it appears that the Chiefs were even more dominant in the passing game than the Ravens were in the running game. On the other hand, the Jets, who, if you remember, place bottom 10 each of these seasons in rushing, had two seasons of passing mediocrity in 2021 and 2022, but outside of that ranked bottom 5 in total passing yards, including ranking dead last in 2020.

Just like with the rushing side, in addition to exploring total passing numbers, I also analysed 'Passer rating', which should better determine the quality of each team's passing game. From this set of graphs, I saw that my previous analysis that the Chiefs absolutely dominated the passing game during this time may not have been completely accurate. While they were still arguably the best, ranking in the top 5 for passer rating, 4 out of the 5 seasons, they fell closer to league average in 2023 after leading the league in 2022. So still very, very good, and probably the best, but not as insane as we may have previously thought, as they were likely atop the league in passing attempts every season, something we will fact-check later.

As far as the Jets go, they are, in fact, just as bad as advertised, ranking in the bottom 5 every season. Even the season they were around league average in total passing, they ranked second-to-last in passer rating.



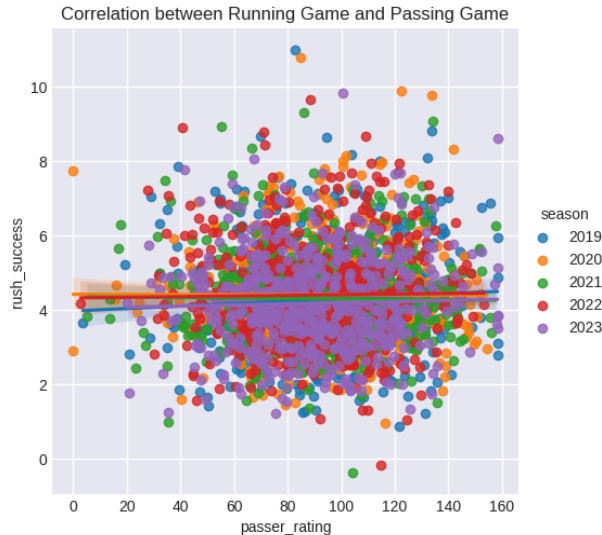
I also want to mention that there is quite a bit of variation in these graphs. As we said before, the average passer rating for the league only had a spread of about 6 points in these 5 seasons, but the teams that are actually good and bad change significantly from year to year. This matches the discrepancy that we saw in the rushing part.

For further analysis, I created tables that displayed each team's ranking for every passing category to give more context about which teams were the best and worst passing teams. From the tables, I can confirm my secondary analysis of the Chiefs' passing game in that, while they were top three every year in passing yards, they were also top three every year in pass attempts as well. In 2023 specifically, the Chiefs' worst ranking in passer rating in these 5 seasons, they ranked 25th in pass average and interceptions. So they were still throwing the ball a lot, but they were not as effective as they were in previous seasons. Their best seasons were 2019, 2020, and 2022, when they ranked in the top ten in passing average, touchdowns, and interceptions, leading to a top 4 ranking in passer rating in each of these seasons. Even in 2021, when they ranked 17th in pass average and 21st in interceptions, they still ranked 6th in passer rating. So overall, very good numbers regardless, but it is important to remember just how much they throw the ball every game to put off those numbers.

The Buccaneers were another team that passed a lot, leading the league in pass attempts from 2020 to 2022. However, this volume of passing only led to barely ranking in the top ten in passer rating in 3 years and outside the top 15 in the other two seasons. In 2022, especially, they threw the ball the most out of any team, but only ranked 16th in passer rating. That team went 8-9, somehow making it to the playoffs, losing in the first round (to the Cowboys of all teams). This was Tom Brady's last season in the league, and he led an offense that was the very definition of air raid. They attempted over 43(!!!) passes a game, which is insane for any QB, let alone a 44-year-old QB. While that formula may have worked in previous seasons (they did win the super bowl in the 2020 season), it definitely was not effective in 2022.

On the other side of things, the 2019 Titans, who we mentioned earlier, had the highest passer rating that season, with only the 5th fewest pass attempts, but they were 2nd in pass average, and had a 3.9 TD to INT ratio, which was only 2nd in the league behind the Chiefs.

The Ravens also only had one season where they didn't rank in the bottom 7 in passing attempts, but still managed to have 2 seasons with rankings in the top ten for passer rating.



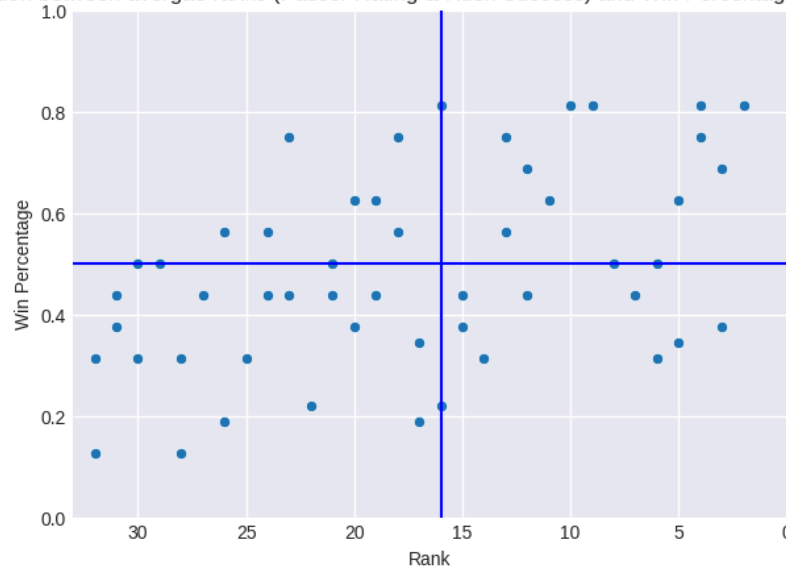
After exploring the rushing and passing data on its own, looking at both the individual teams and the entire league as a whole, I analyzed if there was any correlation between the running game and the passing game using regression plots and models. The short answer is that there is no correlation between the running game and the passing game based on the data. Whether we use the average variables of the success variables, the R^2 value is nearly 0 in both models, and the messy data shown on the scatterplots support this conclusion.

I was initially surprised by the result of these tests, but I think it's important to mention that this is a surface-level test, with little to no context, so this does not mean that we can use this analysis to completely reject the idea that 'the run sets up the pass'. However, the conclusion that we *can* get from these tests is that there is no evidence that a better running game results in a better passing game and vice versa. That actually makes sense, because there are teams like the Buccaneers that we talked about before, that had a good passing game, on average, in each of the five seasons, but were consistently ranked low in the rushing game. And the opposite team was the Ravens, who ran the ball effectively, but also didn't rank highly in the passing game.

After this, I wanted to see if there was any correlation between a good rushing game or a good passing game and winning, considering that winning is the goal of the sport. Shockingly, when I constructed a model that showed the relationship between the combined rank of passing and rushing success and win percentage, the model returned an R^2 value of only 0.30. While there is a clear positive correlation, the results of the model suggest that only 30% of a team's wins can be attributed to their average offensive rank. Again, this was surprising at first, but Offense is only 1/3rd of the game, the other two-thirds including Defense and Special Teams. If I ran this test on the defensive side, I would expect the model to produce a higher R^2 value.

In addition to the regression models, I also created a scatterplot with quadrants that displayed a team's average rank and its corresponding win percentage. Looking at the 1st and 3rd quadrants of the scatterplot, it appears that teams with really bad offensive rankings lead to lower win percentages, more so than teams with really good offensive rankings lead to higher win percentages.

Correlation between average ranks (Passer Rating & Rush Success) and Win Percentage (2019 - 2023)



To conclude my report, here is a summary of my findings in this project:

- There is not a big difference in league rushing or passing data year to year (though there is more variation in the passing data than in the rushing), but there is a big difference in the specific team rankings for both passing and rushing. There are some outliers, like the Chiefs and Ravens being at the top of the passing and rushing game, respectively, and the Jets being at the bottom in both, but overall, the league might not have changed much as a whole from 2019 to 2023, but the individual teams did.

- Just because a team had a lot of rushing or passing yards in a season, does not necessarily mean they were an overall successful team in that area. The number of attempts and turnovers is an important aspect to consider when talking about how effective a team is in a certain area.

- There is no evidence to support a correlation between a good running game and a good passing game. Being successful at one does not equate to success in the other.

- Offense as a whole (passing and rushing together) is only one part of the game of football, so a team with high offense rankings doesn't automatically mean they will have a high win percentage, although it appears that having low offensive rankings leads to fewer wins than having high rankings leads to more wins.

In the future, I would like to explore more years of data and more seasons, so I can explore trends and look at how the league has changed over a longer period of time. That was my initial goal for this project, but with only 5 years of data, there isn't really enough to explore any type of trends.

I would also like to explore the statistics of the defensive side of the ball, as we discovered that the offensive statistics only contribute to about 30% of wins, and I am curious to see how much the defensive statistics contribute.

Lastly, I would like to explore more advanced statistics like EPA (Expected Points Added) per play, and QBR (QB rating), to really get an idea of what teams and what players are generating the most production. Passer rating has its own problems, and 'rush success' is a formula that is derived from passer rating, so I would love to work with some more advanced numbers and formulas.

In this project, I analyzed several different aspects of the sport that can be used as a springboard for more in-depth analysis. This report may be used as a resource to gain more information about NFL statistics and be used as a template to further explore different facets of the game, including Defense and Special teams, and other types of stats. This project really just scratches the surface of the things you can do with NFL data. I look forward to working more with this in the future!

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